MTH316

Examination - January Semester 2025

Multivariable Calculus

Saturday, 19 April 2025 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Define a function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ by

$$f(x, y) = \frac{\sin x}{\sqrt{2x^2 + y^2}}.$$

Question 1a(i)

Determine

$$\lim_{(x,y)\to(\pi/2,1)} f(x, y)$$

if it exists. Otherwise, show that it does not exist.

(2 marks)

Question 1a(ii)

Determine

$$\lim_{(x,y)\to(0,0)} f(x, y)$$

if it exists. Otherwise, show that it does not exist.

(8 marks)

Question 1b

Define a function $g : \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\} \longrightarrow \mathbb{R}$ by

$$g(x, y) = \begin{cases} \dfrac{c(2x - y)}{\sqrt{y} - \sqrt{2x}}, & y \neq 2x \\ \sin c, & y = 2x. \end{cases}$$

Determine all value(s) of c for the function g to be continuous at (0, 0).

(10 marks)

Question 2

An object has a solid region E described by

$$E: \{(x, y, z) | -2x \leq y \leq 2x, -x \leq z \leq x, 0 \leq x \leq 1\}.$$

Question 2a

Sketch the projections of the object onto the three Cartesian coordinate planes. Label the axes intercepts.

(6 marks)

Question 2b

Set up and evaluate a triple integral to compute the volume of the solid.

(8 marks)

Question 2c

If the object is rotated 90° about the x-axis, state the new region E of the object and set up an integral to calculate the volume of the solid. Do not evaluate the integral.

(6 marks)

Question 3

The sound pressure level in decibels (db) over the seating area of an open-air concert is modelled by the function $L_p : \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined below.

$$L_p(x, y) = 70e^{-(x+1)^2} + 50e^{-y^2}.$$

Question 3a

An audio engineer with a microphone walks from (1, 2) along the path $y + 2x = 3$ in the direction of decreasing y. By evaluating the directional derivative, explain if the microphone registers an increase or decrease in the sound pressure level.

(5 marks)

Question 3b

Apply the 1st and 2nd derivative tests to show the classification of each stationary point.

(8 marks)

Question 3c

Determine the absolute highest and lowest sound pressure levels over the line

$$x = 0, -1 \leq y \leq 1.$$

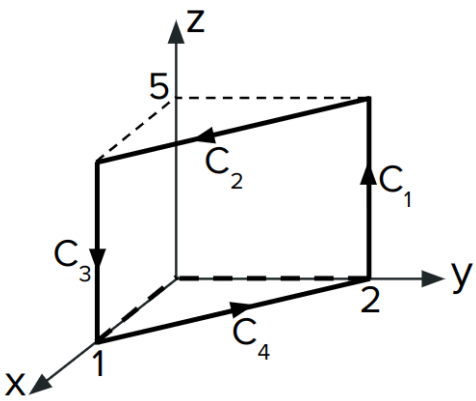
(7 marks)

Question 4

Define a vector field $\mathbf{U} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by

$$\mathbf{U}(x, y, z) = \begin{bmatrix} e^{z/2} \\ x^2 \\ y + 2z \end{bmatrix}.$$

The figure below shows paths C_1 to C_4 that form a rectangle that is perpendicular to the xy-plane.



Question 4a

Compute directly the line integrals of $\mathbf{U}(x, y, z)$ along paths C_1 and C_2.

(10 marks)

Question 4b

Use Stokes' theorem to evaluate the line integral by $\mathbf{U}(x, y, z)$ along the combined path $C_3 + C_4$.

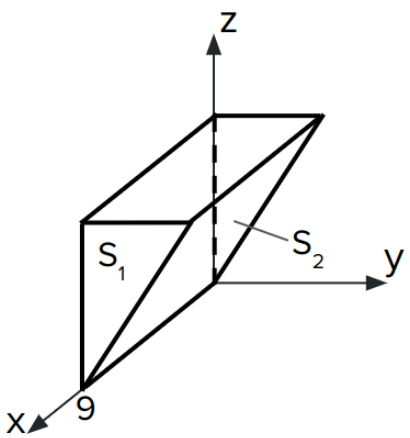
(10 marks)

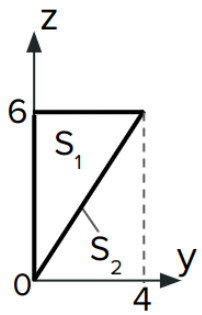
Question 5

Define a vector field $\mathbf{F} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) = \begin{bmatrix} x/3 \\ x - z \\ y^2 \end{bmatrix}.$$

Define a polyhedron in the diagram below, where all five faces of the polyhedron are orientated with outward normal vectors. S_2 is the slanted face of the polyhedron.





Question 5a

Compute the flux of $\mathbf{F}(x, y, z)$ through surface S_1.

(5 marks)

Question 5b

Evaluate the flux of $\mathbf{F}(x, y, z)$ through surface S_2.

(10 marks)

Question 5c

Use the divergence theorem to evaluate the flux of $\mathbf{F}(x, y, z)$ through the other 3 faces of the polyhedron (outside of S_1 and S_2).

(5 marks)

----- END OF PAPER -----